Video 8 – Bias-Variance Trade-off

Video Contents

For SAFE SKIPS: Please read the conclusions in bold.

Video Synopsis

Introduction **04:51 – 11:57**

In this lecture the bias-variance trade-off will be discussed in its entirety as a stand-alone theory regarding generalisation. Learning curves will also be introduced and a comparison will be made with the VC analysis from lecture 7. So far learning has been dealt with as a trade-off between approximation (in-sample) and generalisation (out-of-sample). What this means is that, if you have a complex hypothesis set, you have a good chance of having access to a hypothesis that does very well on the training data (good approximation), but you also have a good chance that you end up with a hypothesis that is sub-optimal on out-of-sample data (bad generalisation) because you have so many hypotheses to choose from.

In the VC analysis, we arrived at the following formula that describes how E_{out} is likely to be linked to E_{in}:

$$E_{out} \leq E_{in} + \Omega$$

In words, E_{out} was estimated as E_{in} plus an extra term that depends on the complexity of the model. The bias-variance trade-off uses another way of quantifying / characterising E_{out}. It sees E_{out} as the sum of two errors:

1. The error due to the (in)ability of the hypothesis set to model the target function accurately.
2. The (in)ability of finding the absolute best hypothesis within the set.

Unlike the VC analysis, we can derive the bias-variance trade-off for real-valued targets (in the VC analysis we limited ourselves to binary targets as otherwise the maths would have been far more complicated). A limitation of the bias-variance trade-off is that we need to assume that the error measure is the squared error.

SAFE SKIP: Decomposition of E_{out} ***11:57 – 28:05***

Suppose we have a set of data sets. Hypothetically, we could obtain E_{out} by evaluating the error on every possible data set D. To decompose E_{out} into the two terms that show the trade-off between approximation and generalisation, we start by defining E_{out} in terms of a squared error and we explicitly recognise that this error will be obtained on a specific set of points, namely D.

$$E_{out}\left(g^{(D)}\right) = E_{\boldsymbol{x}}\left[\left(g^{(D)}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2\right]$$

In words, the out-of-sample error of hypothesis g on a dataset D is equal to the expected value taken over all vectors **x** in D of the squared error between hypothesis and real target function. If we want to obtain the expected out-of-sample error across all potential data sets D:

$$E_D\left[E_{out}\left(g^{(D)}\right)\right] = E_D\left[E_{\boldsymbol{x}}\left[\left(g^{(D)}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2\right]\right]$$

In which we can swap the integration over D and x (expectation uses integration so rules for swapping integrals apply):

$$E_D\left[E_{out}\left(g^{(D)}\right)\right] = E_{\boldsymbol{x}}\left[E_D\left[\left(g^{(D)}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2\right]\right]$$

We will now first decompose the quantity: $E_D\left[\left(g^{(D)}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2\right]$ and for this we will use the notion of an 'average' hypothesis g:

$$\bar{g}(\boldsymbol{x}) = E_D\left[g^{(D)}(\boldsymbol{x})\right]$$

which can be thought of as follows: If you had access to all possible data sets D and obtained the best hypothesis on each individual dataset, you could then test each and every best hypothesis on one point **x** and average the result across all g's. This would be your average hypothesis. Clearly this is not a quantity we can or will realise in practice, but it is useful for the derivation. We will use the average hypothesis by adding and subtracting it to our equation:

$$E_D\left[\left(g^{(D)}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2\right] = E_D\left[\left(g^{(D)}(\boldsymbol{x}) - \bar{g}(\boldsymbol{x}) + \bar{g}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2\right]$$

Which can be rewritten as:

$$E_D\left[\left(g^{(D)}(\boldsymbol{x}) - \bar{g}(\boldsymbol{x})\right)^2 + \left(\bar{g}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2 + 2\left(g^{(D)}(\boldsymbol{x}) - \bar{g}(\boldsymbol{x})\right)\left(\bar{g}(\boldsymbol{x}) - f(\boldsymbol{x})\right)\right]$$

As we are interested in expected values (over D) the cross-term $2\left(g^{(D)}(\boldsymbol{x}) - \bar{g}(\boldsymbol{x})\right)\left(\bar{g}(\boldsymbol{x}) - f(\boldsymbol{x})\right)$ will conveniently disappear: from the perspective of D, $\bar{g}(\boldsymbol{x})$ is a constant as it was already the expected value across all D. The expected value of a constant is the constant itself. The expected value of $g^{(D)}(\boldsymbol{x})$was defined as $\bar{g}(\boldsymbol{x})$, so the cross term is:

$$2\left(\bar{g}(\boldsymbol{x}) - \bar{g}(\boldsymbol{x})\right)\left(\bar{g}(\boldsymbol{x}) - f(\boldsymbol{x})\right) = 0$$

Hence, we are left with:

$$E_D\left[\left(g^{(D)}(\boldsymbol{x}) - \bar{g}(\boldsymbol{x})\right)^2 + \left(\bar{g}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2\right] = E_D\left[\left(g^{(D)}(\boldsymbol{x}) - \bar{g}(\boldsymbol{x})\right)^2\right] + \left(\bar{g}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2$$

where the second term can be taken out of the square brackets as it is a constant. So we now have:

$E_D\left[\left(g^{(D)}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2\right] = \overbrace{E_D\left[\left(g^{(D)}(\boldsymbol{x}) - \bar{g}(\boldsymbol{x})\right)^2\right]}^{1:variance} + \overbrace{\left(\bar{g}(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2}^{2:bias}$ which can be interpreted as follows:

The left-hand side is the average error across all data sets which now has been decomposed into:

1. The expected value across all data sets of the error between the best hypothesis obtained for any one dataset and the 'average' hypothesis. The average hypothesis can be interpreted as a 'best' hypothesis as is it the average of all best hypotheses obtained with all data sets. This term we will call **the variance term.** Compare this term to the definition of variance: $var(x) = E\left[\left(x - E(x)\right)^2\right]$ and you will see that variance is indeed an appropriate term!
2. A term which describes how big the error is with the 'average' ('best') hypothesis and the real target function. This term we will call **the bias term.**

Note that both bias and variance at this stage are still functions of **x.** Reintroducing the averaging across **x** that we dropped previously:

$$E_{\boldsymbol{x}}[\text{bias}(\mathbf{x})+\text{var}(\mathbf{x})] = \text{bias} + \text{var}$$

and the latter are the bias and variance terms that we will continue with. **So we have now shown that Eout can be thought of as having a bias and a variance component.**

A simple thought experiment **28:05 – 30:52**

The trade-off between bias and variance can be investigated with a simple thought experiment. Suppose you have 1 hypothesis in your hypothesis set. In that case, no matter what data set you use, you will always arrive at this one hypothesis. It is likely not very close to the real target function, i.e the bias is high, but it will always be the same, i.e. the variance is low.
Now consider a large number of hypotheses. You are likely to come up with a final hypothesis that is far closer to the real target function, i.e. the bias is low. But this comes at a price: any new data set is likely to result in a slightly different hypothesis, i.e. the variance is high.

Bias-variance trade-off exemplified using a sinusoid. **30:52 - 44:36**

In a further experiment, a sinusoid target function is approximated using two hypothesis sets: H_0 consisting of all horizontal lines (the constant model) and H_1 consisting of all straight lines (the linear model). For H_0 the best you can do is to approximate the sinusoid with a line at height 0. The mean squared error will be 0.5. For H_1 you can do slightly better by choosing a line with a certain slope. This results in a mean squared error of 0.2, so the linear model seems to perform better. When you now train these models repeatedly on two data points and average the resulting hypothesis over all possible data sets of two points, something interesting becomes apparent:

1. The average hypothesis obtained across the two points is the 'best' hypothesis that we found earlier. The resulting error is the bias in our hypothesis.
2. What we also see, is that the variance in best hypotheses for the H_0 model shows far less variation than that for the H_1 model.

Hence the trade-off becomes apparent: simpler models will have higher bias, but lower variance. In this case, when taking into account both bias and variance, the H_0 model beats the H_1 (i.e. the more complex model)!
The important lesson in machine learning is that you will not know the target function. All you have is the data generated by the target function and **you should always decide on your model complexity based on the data you not have and not on how complex you expect the target function to be.**

Learning Curves **46:19 – 50:26**

Learning curves are plots of E_{out} and E_{in} as a function of N, the number of samples available for training. Starting with a learning curve example for a simple model: we can see that this model is too simple as the asymptote that the E_{out} and E_{in} both converge to, lies at quite a high level. We further observe that E_{out} reduces as we have more samples to learn from. This is expected, as a larger number of samples will be able to better represent the out-of-sample situation. For E_{in} we see that the error increases. This is due to the fact that any model will be able to approximate a limited number of samples better than a larger number of samples. For the more complex model, we see similar curves, but with some important differences. We see that E_{in} starts at 0. In other words, for a very small data set, the model is perfect! The largest number of samples for which the model has 0 $E_{in.}$ is equal to the VC dimension of the model. However, for this same number of samples the out-of-sample performance is very bad. In other words, we cannot generalise what we have learned in-sample to the out-of-sample reality. The larger difference between E_{out} and E_{in} is directly related to the complexity of the model, as we know from our VC analysis that a more complex model results in a larger difference (Ω) between E_{out} and $E_{in.}$ We also see that it takes more data for E_{out} and E_{in} to converge, but when they do, the asymptote is at a far lower error level than for the simple model. Hence, these plots show that a more complex model can result in better performance, but only if sufficient data is available for training.

Bias-Variance trade-off versus the VC analysis **50:26 – 54:32**

A comparison of the bias-variance trade-off with the VC analysis shows that, if we plot learning curves for both theories, their conclusions are the same (fortunately ☺). The difference is in how we arrive at these results: the VC analysis starts with the quantification of E_{in} which, after compensating for the complexity of the model Ω, results in E_{out}: $E_{out} \leq E_{in} + \Omega$. In the case of the bias-variance trade-off we had a notional (theoretical) best 'average' hypothesis. The performance of this best hypothesis will be dependent on the complexity of the hypothesis set and the constant error it makes when varying N is called the bias. We see that the bias is the exact same asymptote that E_{out} and E_{in} converged to in the learning curves for the VC analysis. The variance is related to how expressive the hypothesis set is. E_{out} can now be determined by adding the variance to the bias term and we see that the variance term decreases as we have more data. This is expected as, with more data, there will be fewer hypotheses that perform well on all data points. We also see that in the bias-variance trade-off, E_{in} does not play a role at all, like it did in the VC analysis.

The bias-variance trade-off applied to linear regression **54:32 – 1:01:12**

In the last part of this lecture Prof. Abu-Mostafa uses linear regression to generate exact learning curves. To be able to do this symbolically, we use the equations for linear regression as we have already seen them in lecture 3: $\boldsymbol{w} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$, but we add a noise term to the output y:

$$\boldsymbol{w} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T(\boldsymbol{y} + \boldsymbol{noise})$$

We do this so we are able to use a trick to get an out-of-sample error on the exact same data set **X**. The trick is to use the same inputs **X** and outputs **y**, but now with a different noise vector, thus effectively creating a new data set. With this setup, we can mathematically show that the bias term in this case is exactly equal to the noise variance, σ^2, and that the in-sample error is zero as long as the number of data points used for training is less than the VC dimension of the model. We can also quantify by how much the expected in-sample error is better than the bias:

$$\overline{E_{in}} = \sigma^2\left(1 - \frac{d+1}{N}\right)$$

(with the bar indicating expected value) showing that the expected in-sample error is lower than the bias by a fraction determined by the VC dimension of the model and the number of data samples available for training. As the number of data samples increases, E_{in} will converge to the constant bias term equal to σ^2. For the out-of-sample error we can show that:

$$\overline{E_{out}} = \sigma^2\left(1 + \frac{d+1}{N}\right)$$

which is higher than the bias by the exact same term. Hence also E_{out} will converge to the bias term as N increases. The difference between these two terms:

$$\sigma^2\left(1 + \frac{d+1}{N}\right) - \sigma^2\left(1 - \frac{d+1}{N}\right) = 2\sigma^2\left(\frac{d+1}{N}\right)$$

is the generalisation error. It should be noted that this generalisation error is composed of a fraction containing the VC dimension (in this case exact) in the numerator and the number of training samples in the denominator.

In conclusion, this example shows that:

- The bias depends on the model complexity in terms of its ability to 'best' approximate the target function.
- The variance depends on the variation in hypotheses that are the 'best' approximator on any given data set and that this variance reduces with increasing availability of training data.
- The generalisation error is directly related to the model's VC dimension and the number of samples available for training.